AYMANE AIT DADS

AI Video Research Engineer Intern | LLM Fine-Tuning · Computer Vision · Deep Learning Research

Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle competition ($106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) and conducting 20+ single-variable ablation studies to isolate optimal training configurations. Ranked 1st in EURECOM class for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 fine-tuning, and submitted results to the NTIRE 2026 @ CVPR CodaBench international benchmark, demonstrating rigorous academic benchmarking. Core competencies include transformer architecture research, computer vision, NLP, PyTorch, Hugging Face Transformers, and PEFT methods including LoRA and QLoRA. Available for internship in Europe immediately; authorized to work in France with no visa restrictions.

CORE COMPETENCIES

LLM Fine-Tuning | Computer Vision Research | Transformer Architecture | Ablation Studies | Video & Image Understanding | Deep Learning Research | Model Evaluation & Benchmarking | PyTorch & GPU Engineering

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a clustering pipeline using K-Means and Random Forest on **100K+ fiber-optic network records**, mapping customer performance profiles to 5 commercial service tiers adopted by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows built in Python, Scikit-learn, and Pandas.
- Classified network quality across thousands of nodes using upload speed, latency, and jitter features, delivering a **production-ready model** directly consumed by the network optimization team.
- Presented stakeholder-facing Power BI dashboards translating model outputs into actionable maintenance recommendations for **non-technical decision-makers**.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition
Kaggle · In Progress (June 2026) · Top 10% Leaderboard · $106K+ Prize Pool

- Achieved **top 10% public leaderboard (0.80/1.0)** in a $106K+ competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - spanning OOM patching, Mamba CUDA kernel disabling, and Triton ptxas permission fixes - and ran 20+ single-variable ablations confirming completion-only loss, packing strategy, and adapter scale as critical training axes.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - EURECOM ImSecu + NTIRE 2026 @ CVPR
EURECOM Course + NTIRE 2026 @ CVPR CodaBench · 2025 · 1st in Class

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and an ensemble of 3 models with 10-view TTA on 250K images spanning 25+ generator types.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), confirming out-of-distribution generalization as the dominant evaluation axis.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

LLM Fine-Tuning & NLP: LoRA | QLoRA | SFTTrainer | PEFT | Hugging Face Transformers | Unsloth | TRL | vLLM | chain-of-thought | Mamba-Transformer | MoE | prompt engineering | LangChain

Computer Vision & Deep Learning: PyTorch | CLIP | ViT | CNN | DenseNet | OpenCLIP | TensorFlow | FP16/BF16 mixed precision | ablation studies | TTA

Data Science & MLOps: Python | Scikit-learn | Pandas | NumPy | SQL | Git | Linux | CUDA debugging | Triton | Power BI | Docker | PostgreSQL